

Title of the experiment

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1 Introduction

Optical tweezers, first demonstrated by Arthur Ashkin in 1986, use a tightly focused laser beam to trap and manipulate microscopic particles through the gradient force arising from radiation pressure. When a dielectric particle with refractive index greater than the surrounding medium is placed near the focus of a laser beam, the intensity gradient induces a restoring force directed toward the focal point, forming a stable three-dimensional trap. For small displacements from the equilibrium position, this restoring force is linear and the trap behaves as a Hookean spring.

Optical tweezers have become an essential tool in biophysical research, enabling direct measurement of piconewton-scale forces and nanometer-scale displacements relevant to single-molecule mechanics, such as the stepping of molecular motors and the elasticity of DNA. The trap stiffness k depends on the laser power, the numerical aperture of the focusing objective, and the optical properties of the trapped particle.

In this experiment, a single-beam optical tweezers setup is used to trap polystyrene microspheres suspended in deionized water inside a flow cell. By varying the volumetric flow rate of the surrounding fluid, a controlled viscous drag force is applied to the trapped bead.

2 Material and Methods

Instrumentation

A single-beam optical tweezers instrument was used in this experiment. The trapping laser is a 658 nm, 75 mW diode laser, whose beam is directed through two laser-line mirrors designed to reflect light in a narrow band around 633 nm. Between the two mirrors, a convex lens mounted on a sliding rail expands the beam diameter to fill the back focal plane of the microscope objective, maximizing the trapping force. The expanded beam is then focused by a high numerical aperture oil-immersion objective into the flow cell to form the optical trap. Imaging was performed by a CCD camera positioned behind a dichroic mirror, illuminated by a blue LED placed on the opposite side of the flow cell. Laser safety glasses were worn at all times during operation, reducing the laser intensity by at least two orders of magnitude (figure 1).

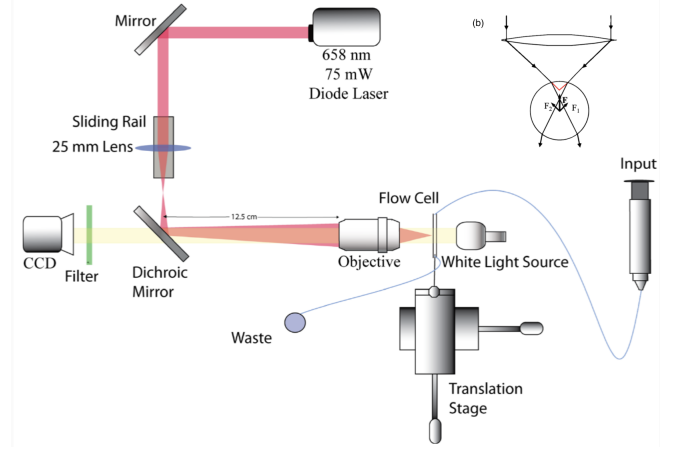


Figure 1: Experimental design of the optical path. (b) Design of the optical cavity of the objective.

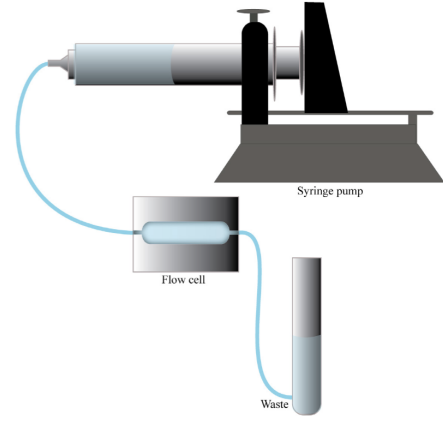


Figure 2: The scheme of flow cell design

Flow Cell Preparation

The flow cell consists of a Plexiglas spacer with two glass cover slips bonded to each side, forming a sealed chamber with an input and output tube (shown in figure 2). Prior to mounting, the cell was filled with deionized water and inspected to ensure all air bubbles were removed from the tubing and valve system. A drop of immersion oil was applied between the objective and the cell to satisfy the optical requirements of the oil-immersion objective. The flow rate of liquid through the cell was controlled by a syringe pump, which allowed precise adjustment of the volumetric flow rate. Measurements were taken at eight flow rate settings: 25, 50, 75, 120, 125, 130, 135,

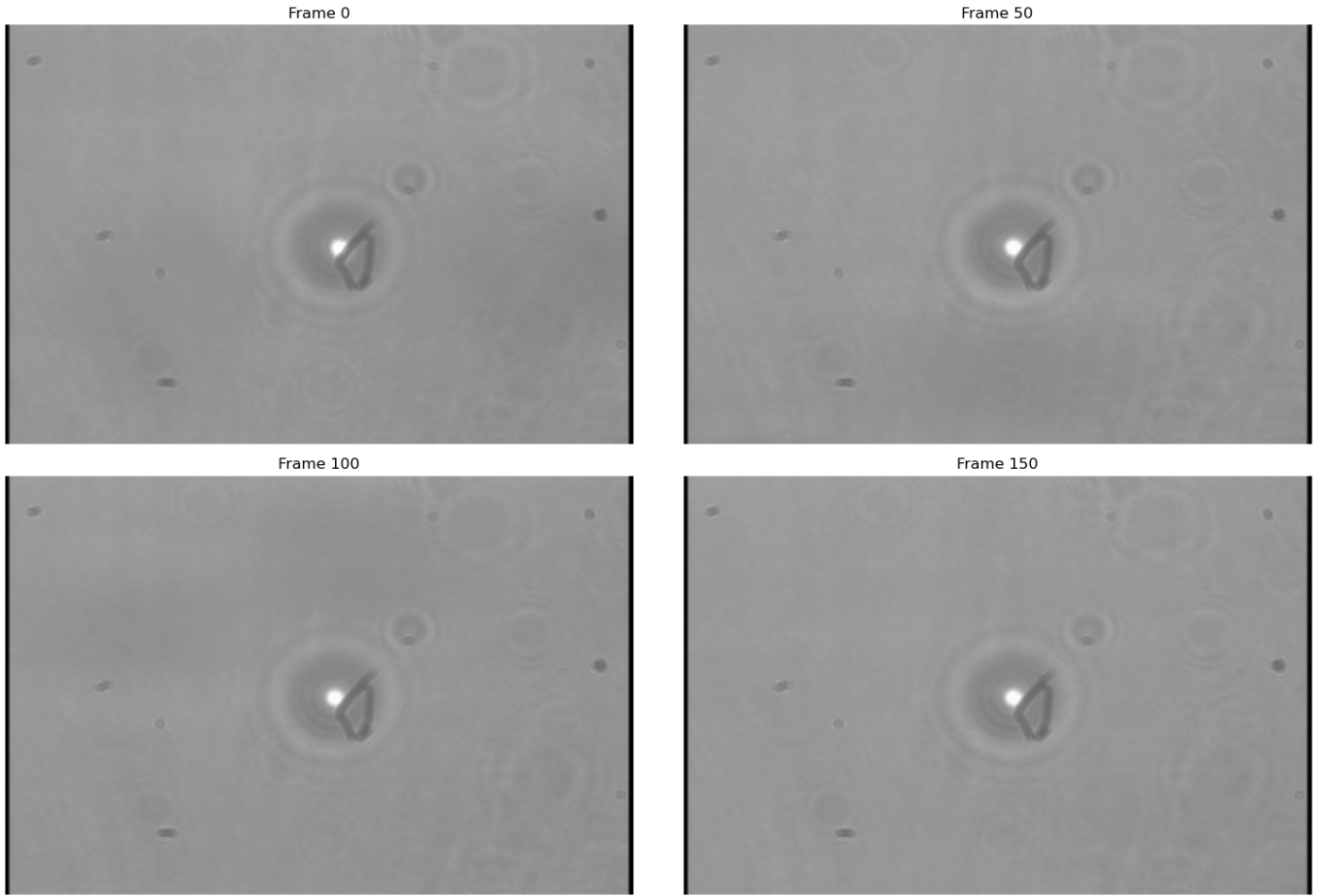


Figure 3: Video snapshot of the bead trapped at 4 different frmaes.

and $145 \mu\text{L}/\text{min}$.

Bead Solution

The trapped particles were polystyrene microspheres with a diameter of $4.16 \pm 0.21 \mu\text{m}$ and index of refraction $n = 1.50$, suspended in deionized water ($n = 1.33$). The refractive index contrast between the bead and the surrounding medium is the physical origin of the trapping gradient force. A small volume of bead solution was introduced into the flow cell once the optical trap was confirmed to be positioned inside the cell.

Data Acquisition

Video data was recorded using Pinnacle Studio 24 via a Pinnacle USB-710 capture device, initially saved as MPEG files and exported for analysis. The CCD camera acquired frames at 30 frames per second at a resolution of 720×480 pixels. For each flow rate setting, three video recordings of a trapped bead were captured. Additionally, for each trapped bead recording, a separate video of a freely flowing, non-trapped bead at the same flow rate was recorded in order to independently measure the local fluid velocity near the focal plane. Freely flowing beads

that were near the focal plane were selected to minimize systematic error arising from the flow velocity gradient across the cell cross-section.

Image Analysis

Video files were analyzed using Python with the OpenCV, NumPy, and SciPy libraries. Each frame was converted to grayscale and a region of interest (ROI) spanning rows 240–270 was selected to isolate the bead. The pixel-to-physical-unit scale was calibrated by measuring the bead diameter in pixels from an intensity profile and applying:

$$\text{pixel size} = \frac{4.16 \mu\text{m}}{\text{bead width (pixels)}} \quad (1)$$

Bead position in each frame was determined by locating the peak of a Gaussian-smoothed ($\sigma = 5$ pixels) intensity profile. Outliers in the position time series were removed using Z-score thresholding (threshold: 2σ), with outlier values replaced by the series median. The range of motion of the trapped bead was computed as the difference between the maximum and minimum positions in the cleaned time series. For freely flowing beads, velocity was computed by tracking the bead centroid across

frames and dividing the displacement by the elapsed time.

Force Calculation and Trap Stiffness

The viscous drag force on a trapped bead was calculated using Stokes' law:

$$F_{\text{drag}} = 6\pi\eta rv \quad (2)$$

where $\eta = 8.9 \times 10^{-4}$ Pa·s is the dynamic viscosity of water at room temperature, $r = 2.08 \mu\text{m}$ is the bead radius, and v is the measured free-bead velocity at each flow rate. The trap stiffness k was determined by fitting a linear model to the measured force-displacement pairs across all flow rates:

$$F = kx \quad (3)$$

The fit was performed using orthogonal distance regression (ODR) to account for uncertainties in both the displacement x and the force F . The maximum trapping force was estimated from the highest flow rate at which stable trapping was observed.

2.1 Savitzky-Golay filter

The Savitzky-Golay filter is a digital smoothing filter based on local polynomial fitting. Rather than applying a simple moving average that blurs out rapid variations, the Savitzky-Golay filter fits a low-order polynomial (typically quadratic or cubic) to a sliding window of data points. At each position, the filter evaluates the fitted polynomial at the center of the window, producing a smoothed value that preserves the underlying shape of the data while attenuating high-frequency noise. A key advantage of the Savitzky-Golay approach is that derivatives can be computed directly from the fitted polynomials, yielding smooth velocity and acceleration estimates without the need for separate differentiation operations. The quality of the filter depends on two parameters: the window length (number of points included in each local fit) and the polynomial order. For this experiment, a window length of 46 frames (≈ 1.5 seconds at 30 fps) and cubic polynomial order (polyorder = 3) were selected to balance noise reduction with preservation of the underlying motion dynamics. This combination effectively smoothed measurement noise while maintaining the temporal resolution needed to accurately identify velocity extrema corresponding to changes in flow rate.

3 Results

For this section, a single data configuration is picked for analysis demonstration. Data and analysis results are both available on Github (URL: <https://github.com/Lemon-Elf/PHYS5318.git>, branch: optical_tweezers) and the analysis method described below were written in a pipeline python script in Github.



Figure 4: Cropped Gray-scaled snapshot of video collected shown at different frames.

The analysis workflow consisted of five main stages: (1) video frame extraction and preprocessing, (2) pixel-scale calibration, (3) bead position tracking, (4) outlier removal and smoothing, and (5) velocity and force calculation. This section describes each stage in detail.

Video files (MPG format) were loaded and processed frame-by-frame using OpenCV (example shown in figure 3). Each frame was extracted from the video stream and converted to grayscale to reduce computational complexity while preserving the intensity information needed for bead localization (figure 4). The CCD camera recorded video at 30 frames per second with a resolution of 720×480 pixels. A region of interest (ROI) spanning rows 240-270 was selected, as this region contained the trapped bead and optical trap center. This ROI allowed for efficient analysis by focusing computation on the relevant portion of each frame.

To convert measured bead positions from pixels to physical units (micrometers), the pixel scale was calibrated using the known diameter of the polystyrene bead. According to the lab specifications, the beads used have a diameter of $4.16 \pm 0.21 \mu\text{m}$.

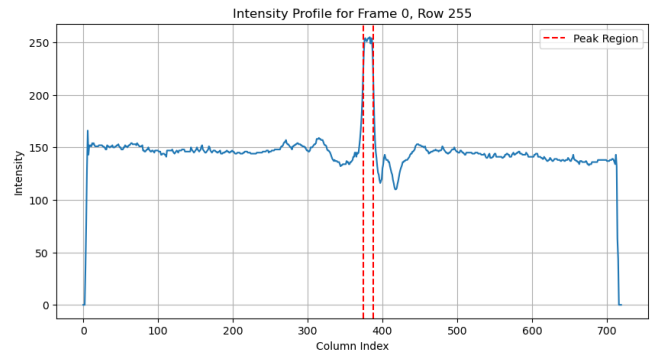


Figure 5: An example intensity profile at a certain frame. Regions within the redline were identified as part of the bead.

The calibration procedure was as follows: An intensity profile was extracted from a single row within the selected region of interest. The background intensity level

was determined as the 25th percentile of the intensity distribution. A threshold intensity was set at 50% of the distance between background and peak intensity. Pixels exceeding this threshold were identified as part of the bead (figure 5). The width of the bead in pixels was calculated as the distance between the first and last pixels exceeding the threshold. The pixel scale was then computed using equation 1.

The uncertainty in pixel size was propagated from the bead diameter uncertainty ($\pm 0.21 \mu\text{m}$) using the same scaling factor. This uncertainty in pixel size subsequently propagates to all measured displacement and velocity values.

For each frame in the video sequence, a one-dimensional intensity profile (picked from row $(270 - 240)/2 = 255$) was extracted from a fixed row within the region of interest. The position of peak intensity in this profile represents the center position of the trapped bead. To improve the precision of peak detection and reduce noise, each raw intensity profile was smoothed using a Gaussian filter with standard deviation $\sigma = 5$ pixels before finding the maximum (figure 6). The maximum intensity position was recorded for each frame, yielding a time series of bead positions with one measurement per frame (30 Hz sampling rate) (top of figure 8).

The raw position time series typically contained spurious outliers caused by noise and temporary loss of bead tracking. Several outlier removal methods were tested and shown in the figure 7: (1) median filtering, (2) interquartile range (IQR) method, (3) Z-score thresholding, and (4) a combination approach using median filtering followed by Savitzky-Golay smoothing.

The Z-score method was selected for the primary analysis due to its robust statistical basis. In this method, points more than 2 standard deviations away from the mean were identified as outliers. Outlier values were replaced with the median of the position time series to preserve the time series length while removing erratic measurements. This approach effectively removed spike artifacts while maintaining the overall motion characteristics of the trapped bead.

Using the cleaned position data (in pixels), displacement metrics were calculated: (1) Average frame-to-

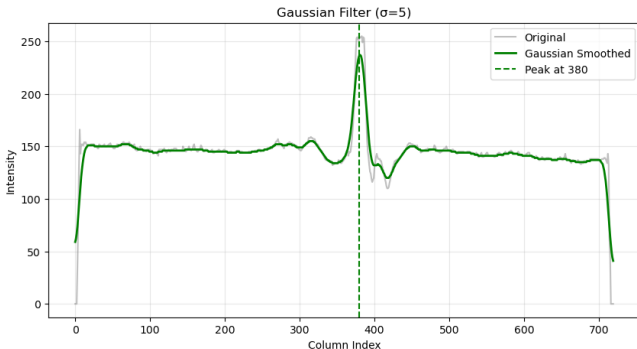


Figure 6: The gaussian filter of the intensity profile with $\sigma = 5$

frame displacement: The magnitude of position change between consecutive frames, averaged over all frames. (2) Total displacement: The net change in position from the start to the end of the recording. (3) Range of motion: The difference between maximum and minimum positions observed.

All displacement values were converted from pixels to micrometers using the calibrated pixel scale. Uncertainties in these quantities were propagated from the pixel-size uncertainty using:

$$\sigma_{\text{displacement}} = |\text{position (pixels)}| \times \sigma_{\text{pixel size}} \quad (4)$$

Velocity was calculated from the position time series as the time derivative of position. To obtain smooth velocity estimates and accurate derivatives, a Savitzky-Golay filter was applied to the cleaned position data with a window length of 46 frames (1.5 seconds) and polynomial order 3. This filter simultaneously smooths the data and enables accurate computation of the first derivative, yielding the instantaneous velocity at each time point.

The Savitzky-Golay velocity estimate was computed as:

$$v(t) = \left. \frac{d(\text{position})}{dt} \right|_{\text{Savitzky-Golay fit}} \quad (5)$$

The resulting velocity time series represents the flow velocity experienced by the trapped bead and is proportional to the syringe pump flow rate setting. Velocity uncertainty was similarly propagated from pixel-size uncertainty:

$$\sigma_v = \frac{|\text{velocity (pixels/sec)}| \times \sigma_{\text{pixel size}}}{\text{time per frame}} \quad (6)$$

The negative velocity minima in the derived velocity curve were identified as the points of maximum flow resistance or beam intensity modulation. These minima were detected using peak-finding algorithms applied to the inverted velocity curve. A discontinuity check was implemented to reject spurious minima that appeared as isolated artifacts rather than genuine physical features. The selected velocity minimum was used to quantify the flow-induced force on the bead at each pump setting (figure 8).

4 Discussion

The optical trap stiffness was determined from the linear relationship between bead displacement and applied force. Figure 9 presents the force versus displacement data, where the applied force was calculated from the viscous drag on the bead as it moves with the fluid flow. The bead displacement represents the equilibrium position shift relative to the trap center when the force is applied.

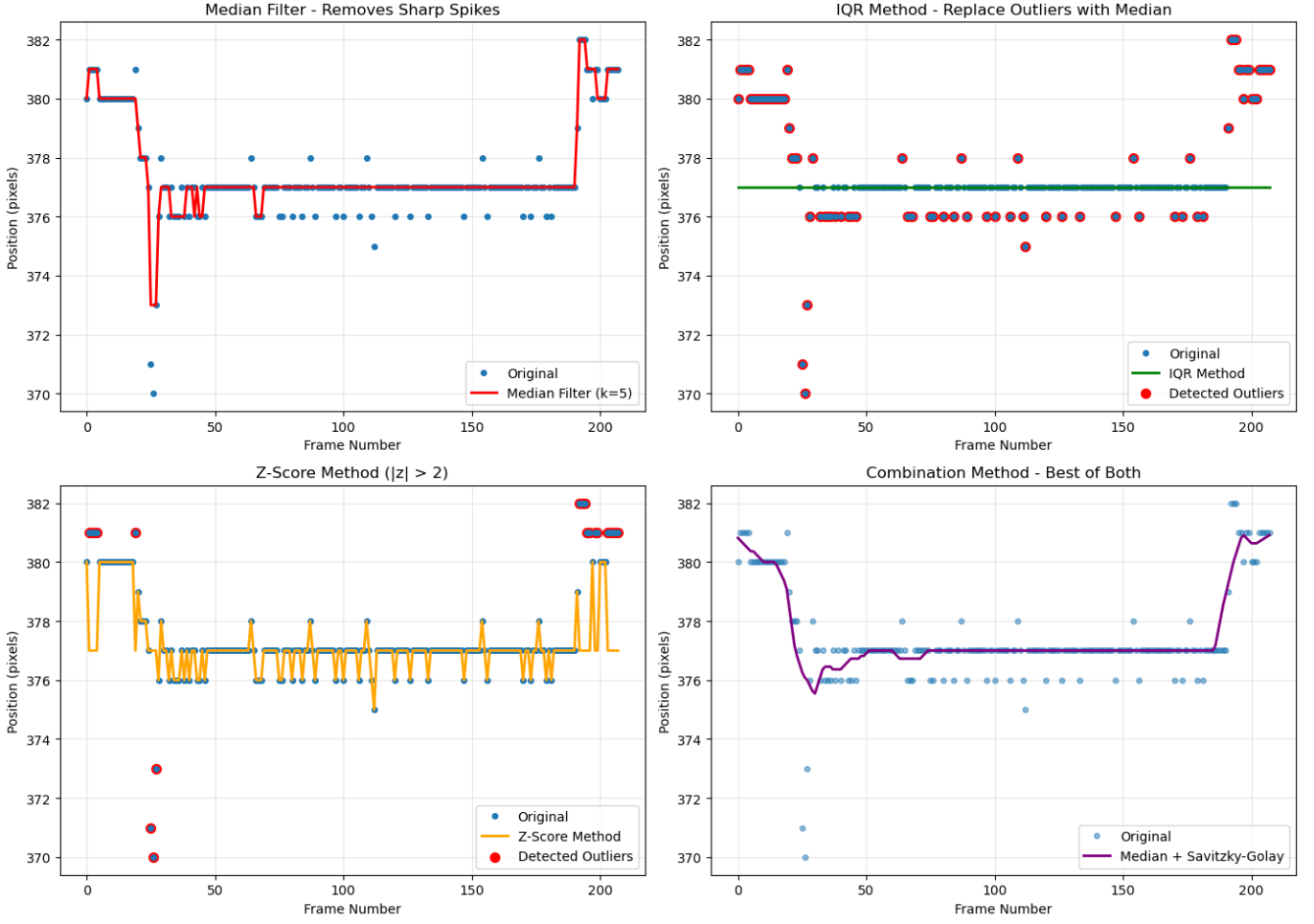


Figure 7: Presentation of efficiency of different filters for the position data points in order to determine the range of positional motion

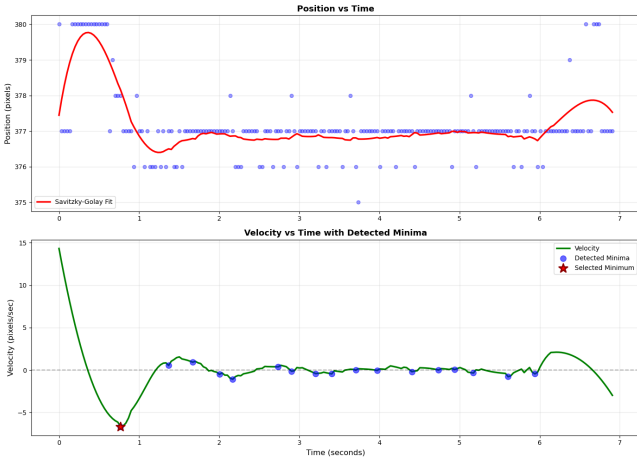


Figure 8: Caption

An orthogonal distance regression (ODR) fit was applied to the force-displacement data, accounting for uncertainties in both coordinates. The fitted trap stiffness was determined to be:

$$k = 0.032 \pm 0.003 \text{ pN}/\mu\text{m} \quad (7)$$

with a coefficient of determination $R^2 = 0.713$. This linear relationship indicates that the trap follows Hooke's law over the measured range of displacements (0 to $\sim 3 \mu\text{m}$). The scatter in the data, reflected in the $\pm 1\sigma$ error band shown in red, is consistent with measurement uncertainties in both position determination from pixel analysis and flow velocity estimation. The trap stiffness value is comparable to typical optical tweezers systems operating at similar laser power and focusing geometry.

The displacement measurements as a function of flow rate are presented in Figure 9 (left). Mean bead displacement increased with flow rate from approximately $0.7 \mu\text{m}$ at $25 \mu\text{L}/\text{min}$ to $2.0 \mu\text{m}$ at higher flow rates, demonstrating that larger flow-induced forces push the bead further from the trap center. The error bars represent the standard error of the mean calculated from three independent measurements at each flow rate setting. At the highest flow rates tested (around $140 \mu\text{L}/\text{min}$), the displacement approached $2 \mu\text{m}$, indicating that the trapping force was approaching its maximum capability before bead escape became probable.

The drag force as a function of flow rate is shown in Figure 9 (right). The mean drag force increased from approximately 0.025 pN at the lowest flow rate to a max-

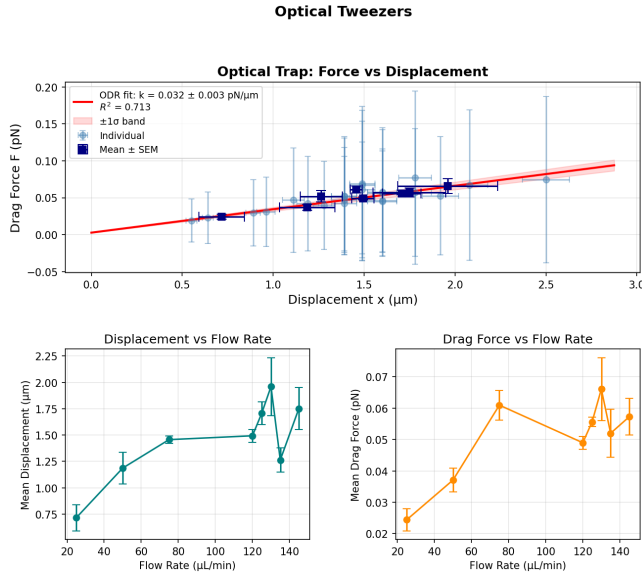


Figure 9: Result of the analysis from the all the data obtained.

imum of approximately 0.065 pN at 60 $\mu\text{L}/\text{min}$, followed by a slight decrease and stabilization at higher flow rates. This data demonstrates the relationship between syringe pump settings and the actual hydrodynamic force experienced by the bead. The uncertainties in measured drag force (shown as error bars) reflect both the variability in flow rate measurement and the propagation of uncertainties through the viscous drag calculation. The maximum drag force measured (~ 0.065 pN) corresponds to the approximate maximum trapping force that can be sustained before bead escape occurs, setting the upper limit of forces measurable with this optical tweezers configuration.

These results validate the use of optical tweezers as a precision force measurement instrument capable of resolving piconewton-scale forces and sub-micrometer displacements, suitable for single-molecule mechanical experiments.

5 Conclusion

In this experiment, a single-beam optical tweezers instrument was calibrated by measuring the displacement of a trapped polystyrene microsphere under controlled viscous drag forces at eight different flow rate settings. Video analysis using Python and OpenCV was used to extract bead positions frame-by-frame, with Gaussian smoothing and Z-score outlier removal applied to improve tracking fidelity. Fluid velocities were independently measured from freely flowing beads to compute the applied drag force via Stokes' law.

An orthogonal distance regression fit to the force-displacement data yielded a trap stiffness of:

$$k = 0.032 \pm 0.003 \text{ pN}/\mu\text{m} \quad (8)$$

with a coefficient of determination $R^2 = 0.713$, confirming a linear (Hookean) response over the measured displacement range of 0–3 μm . The maximum sustained trapping force was approximately 0.065 pN, consistent with the laser power and focusing geometry used. These results are in line with values reported for comparable optical tweezers configurations.

The primary sources of uncertainty in this experiment include the pixel-scale calibration derived from the bead diameter, variability in free-bead velocity measurements, and potential systematic error from the flow velocity gradient across the cell cross-section. Future improvements could include using a position-sensitive detector (PSD) for higher-resolution displacement measurement and incorporating a more controlled flow geometry to reduce velocity uncertainty. Nevertheless, the results demonstrate the ability of optical tweezers to resolve piconewton-scale forces and sub-micrometer displacements, validating their use as precision instruments for single-molecule mechanical studies.

References

- [1] *Experiment 6: Optical tweezers: Measuring piconewton forces*, (Northeastern University, 2026)
- [2] *Savitzky-Golay Smoothing and Differentiation Filter*, (Neal B. Gallagher, Eigenvector Research Incorporated)